Volt VAR Droop Curve Design in Transmission Systems

Voltage and reactive power droop should be designed as part of a coordinated transmission control system. A curve that gives the desired reactive response at one operating point may exhaust reactive reserve, produce unsuitable sharing, or interact with other controllers after a network outage. The central design task is to select a realizable voltage–reactive power characteristic and demonstrate acceptable behavior across network conditions, equipment limits, and control modes.

This report addresses transmission-connected synchronous generators, synchronous condensers, inverter-based resource plants, static VAR compensators, STATCOMs, and voltage-source-converter HVDC terminals. It develops equations, worked examples, a tuning workflow, modelling requirements, and research priorities. The regulatory discussion compares Australia, the European Union, and North America. Numerical examples are analytical illustrations, not recommended settings for a particular connection.

The engineering recommendation is to specify droop in physical units and on explicit per-unit bases, retain a clear distinction between normal voltage regulation and fault response, and optimize settings against both voltage security and dynamic performance. Reactive reserve and coordination with slower controls should be explicit design objectives. Equipment-specific studies remain necessary before selecting numerical settings.

1 Scope and evidence

Transmission Volt/VAR control regulates voltages in an electrically coupled, usually meshed system. The regulated point may be a generator terminal, transformer high-voltage bus, plant point of connection, or an agreed remote bus. These locations are not interchangeable: network impedances, transformer taps, and internal reactive losses alter the relationship between a controller's measured quantities and the support delivered to the transmission system.

NERC's 2018 performance guideline recommends coordinating reactive droop where multiple resources regulate the same or electrically nearby buses. It also illustrates that the same physical characteristic can acquire different percentage labels when the reactive range is normalized differently. That guidance supports coordinated design, but is not a universal numerical setting or a substitute for applicable reliability standards.[1]

The evidence falls into three categories. Grid rules and standards define obligations within their scope; operator reports and modelling guides describe practical requirements and observed failures; research papers establish results under stated mathematical assumptions and test systems. The derivations and proposed design procedures below are engineering analysis. Their assumptions are stated alongside the equations so that a simplified result is not mistaken for a transmission-wide stability guarantee.

Distribution Volt/VAR curves and microgrid Q–V droop are useful conceptual references. However, transmission applications add meshed-network coupling, high-power converters, remote voltage regulation, synchronous excitation systems, and contingency-driven changes in system strength. IEEE 2800-2022 explicitly addresses IBRs connected to transmission and sub-transmission systems and covers reactive control and

1 NERC, Inverter-Based Resource Performance Guideline, September 2018. Chapter 3, especially printed page 28. Historical technical guideline.

abnormal-voltage support. Its public scope should not be confused with distribution interconnection requirements.[2]

2 Definitions and curve conventions

2.1 Sign conventions and bases

Let Q be reactive power injected by the controlled resource into the network: Q > 0 denotes injection and Q < 0 denotes absorption. Let V be the positive-sequence fundamental-frequency voltage magnitude at the specified measurement point. Define v = V/Vbase and q = Q/Qbase. Qbase is an explicitly declared MVAr normalization; it need not equal the plant's MVA rating or either directional capability limit.

For an unsaturated curve without a deadband, use:

$$v = vref - r(q - q0)$$

$$qcmd = q0 + (vref - vm)/r$$

Here r > 0 is droop in pu voltage per pu reactive power, vm is measured per-unit voltage, vref is the reference voltage, and q0 is a reactive bias. A falling measured voltage commands increasing reactive injection. The reactive response gain is K = 1/r, so smaller numerical droop means greater gain. "Steeper" is ambiguous unless the plot axes are specified: Q plotted against V has slope −K, while V plotted against Q has slope −r.

In physical units, define m = rVbase/Qbase in kV/MVAr when Vbase is in kV. Then V = Vref − m(Q − Q0), and the magnitude of the physical reactive gain is G = 1/m in MVAr/kV. If Qbase changes while the physical curve remains fixed, r changes proportionally to Qbase. A controller setting should therefore include the equation and bases, not merely a percentage.

2.2 A base conversion example

Assume Vbase = 220 kV, Qbase = 100 MVAr, r = 0.04, and Q0 = 0. A 100 MVAr increase corresponds to an 8.8 kV decrease along the specified characteristic. The physical slope is 0.088 kV/MVAr and the gain is 11.36 MVAr/kV. At an externally imposed voltage of 0.99 pu with vref = 1.00 pu, the unconstrained command is +25 MVAr.

On a 200 MVAr base the same characteristic has r = 0.08, or 8%. It has not become a weaker physical controller. Similarly, a symmetric sweep from −100 to +100 MVAr covers 0.08 pu voltage, although the one-sided 100 MVAr normalization gives 4% droop. Percentages are comparable only after reconciling voltage base, reactive base, one-sided versus full-span convention, measurement location, and whether deadband is excluded from the slope calculation.

2.3 Reactive current droop

Some controllers use reactive current rather than reactive power feedback. With compatible power-invariant per-unit bases and a sign convention chosen for positive reactive injection, q = v iq. Hence a linear iq–v law is generally nonlinear when expressed as q–v. Near v = 1 pu the two may approximate each other; during substantial voltage deviations their difference matters.

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2 IEEE Standards Association, IEEE 2800-2022 Standard for Interconnection and Interoperability of Inverter-Based Resources Interconnecting with Associated Transmission Electric Power Systems, 22 April 2022. Official catalogue scope and status; full normative text is not reproduced.

For example, iq = 0.5 pu produces q = 0.5 pu at v = 1.0 pu but only q = 0.4 pu at v = 0.8 pu. A constant Q command of 0.5 pu would instead require iq = 0.625 pu at 0.8 pu. Software and OEM documents may define the q-axis current with the opposite sign; verify the transformation and current direction before implementing this relation.

3 Deadbands saturation and curve geometry

3.1 A continuous asymmetric characteristic

Let e = vref − vm. Let dL and dH be nonnegative voltage-error thresholds for undervoltage and overvoltage response. Let KL and KH be positive reactive gains. A continuous curve is:

$$qraw = q0 + KL(e - dL) \qquad when\ e > dL$$

$$qraw = q0 \qquad when\ -dH \le e \le dL$$

$$qraw = q0 + KH(e + dH) \qquad when\ e < -dH$$

$$qcmd = min(qmax,\ max(qmin,\ qraw))$$

This is a non-step deadband: response starts continuously from q0 at each edge. It avoids commanding a finite jump merely because a noisy measurement crosses the threshold. Injection and absorption slopes can differ, as can the available limits. With qmin < q0 < qmax, the saturation voltages are:

$$vlow = vref - dL - (qmax - q0)/KL$$

$$vhigh = vref + dH + (q0 - qmin)/KH$$

These equations expose the design tradeoff. A wider deadband consumes voltage range before support begins; a larger gain reaches capability sooner. For a required injection qreq at voltage vreq below the deadband, KL must be at least (qreq − q0)/(vref − vreq − dL), provided the denominator is positive and qreq is physically available. This is a static lower bound on gain, not a dynamic tuning result.

3.2 Deadband is an operating decision

Choose deadband against measurement uncertainty, routine voltage variation, permitted regulation error, and the minimum disturbance that should trigger support. A deadband selected only to silence noisy Q measurements can conceal poor sensing or unsuitable filtering. Conversely, forcing every resource to respond to every tiny fluctuation can increase reactive circulation and control activity.

Hysteresis and deadband solve different problems. Deadband defines a range of no incremental response; hysteresis makes switching thresholds depend on the previous state. If hysteresis is used, specify its width, memory state, reset behavior, and treatment in simulations. A memory-dependent controller cannot be reproduced by a single static curve alone.

3.3 Bias and voltage schedule changes

The bias q0 reserves a nonzero operating point while preserving a local droop response. In the linear region, changing q0 by Δq0 is equivalent to changing vref by rΔq0. This equivalence breaks across deadbands, saturation, and separate rate limits. Avoid allowing two supervisory functions to independently adjust equivalent quantities without coordination.

Define whether a voltage schedule change translates the complete curve, shifts only the zero-reactive intercept, or resets an internal integrator. Also define how an unavailable unit changes the aggregate capability and response gain. Rescaling Qbase automatically with online capacity changes the plant's physical gain unless the implementation compensates for it.

4 Reactive capability and equipment differences

4.1 Feasibility at the connection point

A useful initial converter bound is $P^2 + Q^2 \le Smax^2$ at rated voltage and within the assumptions of an apparent-power limit. It is not a complete plant capability curve. For a current-limited converter with compatible per-unit quantities, a more relevant instantaneous bound is $p^2 + q^2 \le (v\ imax)^2$. If $|p|$ exceeds v imax, the requested active output is itself infeasible and a priority or curtailment mechanism must intervene.

At a fixed p = 0.9 pu and imax = 1.0 pu, the current-circle reactive capability is approximately 0.436 pu at v = 1.0 pu. At v = 0.95 pu it falls to approximately 0.304 pu. This simple calculation shows why clipping a curve against a voltage-independent Q limit can exaggerate support. Actual converter limits also depend on internal voltage, thermal state, DC conditions, and the implemented current allocation.

Compute net connection-point capability by modelling internal equipment. A useful accounting identity is QPOC = sum(Qunits) + Qshunts − Qinternal, with consistent injection signs. Qinternal includes reactive consumption in series impedances; capacitive cables and filters must be represented with their actual voltage dependence. The aggregate limit is the feasible result of internal network constraints, not always the sum of independent nameplate ratings.

4.2 Comparison of resource types

Resource	Dominant design issue	Consequence for droop studies
Synchronous generator	Excitation and machine capability limits	Include AVR, stabilizer, excitation limiters and terminal constraints
Synchronous condenser	Excitation limits and transformer coupling	Check dynamic reserve and interaction with nearby voltage controls
GFL wind solar or battery plant	Current limits and plant controller architecture	Represent PLL, unit controls, delay and allocation logic
GFM converter	Internal voltage regulation and current limiting	Check interaction of Q–V law, virtual impedance and limiter recovery
SVC	Bounded shunt susceptance	Available MVAr declines with voltage squared at the susceptance limit
STATCOM	Converter current and voltage limits	Current-limited MVAr approximately scales with voltage
VSC HVDC terminal	AC reactive support shares converter capability	Coordinate AC voltage control with active transfer and DC control

For an SVC at a fixed susceptance limit, $Q = BV^2$. For an ideal current-limited STATCOM, reactive output magnitude is approximately $\sqrt{3}$ V Iq using line-to-line RMS voltage. Thus, at 0.8 pu voltage, equivalent 100 MVAr nominal limits yield roughly 64 MVAr for the susceptance-limited case and 80 MVAr for the current-limited case. These are scaling illustrations; converter voltage limits, protection and overload ratings can alter actual performance. Hitachi Energy describes STATCOMs as continuously controlled VSC-based reactive devices that can also be combined with fixed or switched shunts.[3]

4.3 Reserve and priority

Define upward reactive reserve as Rplus = Qmax − Q and downward reserve as Rminus = Q − Qmin, evaluated at the relevant operating state. High voltage regulation gain may consume reserve during ordinary operation, leaving little incremental capability after an outage. A design objective should distinguish economical steady support from fast contingency reserve.

Specify the response when active and reactive commands compete for current. Reactive priority can require active-power reduction; active priority can prevent the requested voltage support. Batteries must be checked in charging, discharging, zero-power and unavailable states. Wind and solar plants require explicit confirmation of reactive operation at low or zero production rather than an assumption that the entire installed fleet remains available.

5 Transmission network sensitivity and sharing

5.1 Deriving a local voltage sensitivity

For small deviations about a solved AC operating point, partition the power-flow Jacobian into angle and magnitude terms:

$$\Delta P = JP\theta \ \Delta\theta + JPV \ \Delta V$$

$$\Delta Q = JQ\theta \ \Delta\theta + JQV \ \Delta V$$

With fixed active injections and an invertible reduced angle block, eliminating Δθ gives ΔQ = JR ΔV, where JR = JQV − JQθ JPθ^{-1} JPV. Then H = JR^{-1} maps incremental reactive injections to voltage changes on the selected independent buses. A slack/reference treatment, fixed controls and consistent units are required. Near singularity, this local approximation becomes ill-conditioned and needs particular care.

This derivation is preferable to assuming that only a resource's own bus matters. Off-diagonal entries describe the effect of one plant on another regulated bus. Recompute sensitivities for credible outages, dispatch patterns and relevant discrete-device states. Once a generator reaches its Q limit, the active equations change and the old sensitivity no longer describes the same control regime.

5.2 Closing the static droop loop

Write the incremental network as Δv = h + HΔq, where h is the voltage disturbance with controlled reactive injections held fixed. In an active linear droop region, Δq = −KΔv, with K diagonal if each controller measures its own selected voltage. Substitution gives:

$$(I + HK) \ \Delta v = h$$

3 Hitachi Energy, STATCOM, Undated. Manufacturer description of VSC reactive compensation and shunt combinations.

$$\Delta q = -K(I + HK)^{-1}h$$

For a scalar sensitivity H > 0, increasing K reduces the final voltage deviation. It does not establish acceptable transient performance, and the formula must be re-solved if a limit or deadband changes the active region. Remote measurements can require a non-diagonal control mapping; replacing that mapping by a diagonal approximation can hide important coupling.

An illustrative scalar case uses H = 0.08 pu voltage per pu Q and h = −0.02 pu. With K = 25, Δv = −0.00667 pu and Δq = 0.1667 pu. The voltage is partly restored by reactive injection. If the available increment is only 0.10 pu, the linear result is infeasible; at that limit Δv = −0.02 + 0.08 × 0.10 = −0.012 pu. Curve design and reserve sizing are therefore inseparable.

5.3 Sharing at a common bus

If two unsaturated resources see the same voltage deviation and have identical reference and deadband treatment, their incremental MVAr contributions are proportional to Qbase,i/ri. Equal per-unit droop produces proportional sharing only when the chosen reactive bases represent the intended participation weights. Different references can create circulating reactive power even before the external disturbance occurs.

For resources A and B with reactive bases of 100 and 50 MVAr and r = 0.04, an externally imposed 0.01 pu voltage decrease commands 25 and 12.5 MVAr. The 2:1 response follows from the declared bases. If B saturates at 8 MVAr, that sharing ratio cannot persist. Whether A supplies additional MVAr depends on the resulting network voltage and its own remaining capability; it is not guaranteed to replace all of B’s missing response.

At distinct buses, local voltages differ, so equal slopes cannot generally impose exact system-wide sharing. Abdolmaleki, Simpson-Porco and Bergna-Diaz develop a distributed controller that prioritizes voltage containment and shares reactive power among resources whose voltage constraints are inactive. Their stability analysis includes voltage–angle coupling and saturation, but the reported demonstrations use low- and medium-voltage microgrids. It is relevant methodology, not direct validation for a meshed transmission system.[4]

6 Controller architecture and coordination

6.1 Implementing permanent droop

A controller can track qcmd from the static curve through an inner reactive regulator. Alternatively, a voltage regulator can act on a compensated error ecomp = vref − vm − r(qm − q0). An integral controller driving ecomp toward zero preserves the desired steady droop relation. An integrator acting on uncompensated vref − vm instead tends to eliminate the steady voltage deviation, potentially overriding the intended permanent droop.

Draw the full signal path: voltage sensor, positive-sequence extraction, filtering, droop or compensation calculation, regulator, limiter, communication channel, unit allocation and resource response. Identify where the measured Q enters and whether it is unit output or connection-point exchange. Specify initial states and anti-windup, especially when the commanded Q differs from achievable Q.

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4 Babak Abdolmaleki, John W. Simpson-Porco and Gilbert Bergna-Diaz, Distributed Optimization for Reactive Power Sharing and Stability of Inverter-Based Resources Under Voltage Limits, 28 February 2023, arXiv v2. Sections II–V. Microgrid demonstrations; transmission transfer requires validation.

The WECC 2015 PV validation guide distinguishes plant voltage and reactive-power modes and explains REPC_A choices for droop versus line-drop compensation. It is useful for understanding generic-model architecture. Its flag guidance is historical: the actual simulation platform, model revision and OEM implementation must be checked before transferring parameter values.[5]

6.2 Transformer and remote compensation

Ordinary positive droop relaxes voltage regulation as reactive injection increases. Line-drop compensation estimates or counteracts a voltage difference between measurement and regulation locations. They can have opposite effective signs. For a simplified transformer relationship $v_t \approx v_{poc} + xq$, terminal droop $v_t = v_{ref} - rq$ gives $v_{poc} \approx v_{ref} - (r + x)q$. This illustration assumes near-unit voltage, compatible bases and negligible resistance/angle effects.

Compensation that offsets the xq term can recover a desired high-side behavior, but overcompensation can reduce effective droop or reverse its sign. AEMC's 2018 determination explicitly distinguished reactive-current compensation from droop when retaining both concepts in the rules. This is a practical reason to verify the sign from a physical perturbation rather than infer it from an OEM parameter name.[6]

For a remote regulated bus, include communication delay, stale-data detection, fallback measurements, transfer logic and the contingency that separates the resource from that bus. Test the inferred remote voltage against an AC solution across P, Q and tap positions. A compensation coefficient accurate at one operating point need not remain accurate throughout the operating envelope.

6.3 Coordination across time scales

Use local and plant controls to deliver the specified fast response; use supervisory control to restore schedules and manage reactive reserve; coordinate taps and switched shunts with that response. Time-scale separation is a design hypothesis to test, not a rule that is satisfied merely by labelling one loop "slow." The complete cascade should meet the required response while maintaining acceptable margins.

Avoid a plant regulator that constantly cancels unit-level voltage support. At a hybrid site, allocate requests using available directional headroom and internal voltage constraints. Reallocation should be smooth when a resource reaches a limit or disconnects. Include internal collection-system voltages as monitored constraints even when the transmission connection-point target is met.

For GFM resources, the Q–V relationship often changes an internal voltage reference; a GFL implementation typically produces reactive current or power commands through a grid-synchronized control system. Identical static slopes do not imply identical dynamics. The NREL grid-forming roadmap discusses voltage/reactive sharing and overlapping inverter and synchronous-excitation dynamics, and cautions that microgrid coordination approaches face additional challenges on bulk systems.[7]

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5 WECC, Central Station Photovoltaic Power Plant Model Validation Guideline, 17 June 2015. Section 3.2.1. Historical REPC_A mode and compensation guidance.

6 AEMC, Generator Technical Performance Standards Final Determination, 27 September 2018. Printed page 131, droop and reactive current compensation.

7 Yashen Lin and coauthors, NREL, Research Roadmap on Grid-Forming Inverters, November 2020. NREL/TP-5D00-73476; voltage-control discussion, printed page 19.

7 Dynamic stability and weak network conditions

7.1 A transparent reduced model

Consider a scalar first-order reactive response with time constant τ, no delay, no saturation, and the local network $\Delta v = H\Delta q + h$. Let $\tau\, d\Delta q/dt = -\Delta q - K\Delta v$. Substitution yields a closed-loop pole $s = -(1 + KH)/\tau$. For $K > 0$ and $H > 0$, this simplified model is stable and becomes faster as gain increases. It cannot support the blanket claim that a large reactive gain is always unstable.

Now include a pure measurement/communication delay T in the voltage feedback. The characteristic equation becomes $\tau s + 1 + KH \exp(-sT) = 0$. A stability boundary at $s = j\omega$ satisfies $1 + KH \cos(\omega T) = 0$ and $\tau\omega - KH \sin(\omega T) = 0$. Delay introduces phase loss, so the gain that improves static regulation can reduce stability margin. Additional filters, PI dynamics, PLLs and inner current controls create further modes.

For a vector first-order approximation with a common τ and no delay, the state matrix is $-(I + KH)/\tau$. Stability requires the eigenvalues of $I + KH$ to have positive real parts. A separate sampled direct-update algorithm, $\Delta q[k+1] = -KH\Delta q[k] - Kh$, converges only if the spectral radius $\rho(KH) < 1$. These are different models: an iterative convergence condition must not be presented as a universal physical-controller stability criterion.

7.2 System strength is more than one number

For an approximately inductive Thevenin equivalent on a consistent base, reactance and inverse short-circuit strength give useful intuition about voltage sensitivity. However, a short-circuit ratio does not describe all converter interactions, impedance angle, nearby resource controls, or frequency-dependent network resonances. Use it to screen cases and characterize the study envelope rather than certify stability.

Evaluate weak and strong networks. Weak conditions can increase voltage sensitivity and synchronization difficulties; strong conditions can move control crossover or expose other fast interactions. Active dispatch changes converter headroom and operating-point linearization. Nearby resources switching control modes can change effective network behavior even when the short-circuit level is unchanged.

7.3 Validation hierarchy

Begin with AC power flow and contingency analysis to reject infeasible curves. Use linearization and eigenvalue or frequency-response analysis to identify interacting modes and tune margins. Use positive-sequence dynamic simulation for broad operating-case coverage when the model is suitable. Escalate to detailed electromagnetic transient models for fast converter effects, unbalance, limiter transitions, phase jumps, or interactions that the positive-sequence representation cannot resolve.

AEMO's current modelling requirements page identifies Power System Model Guidelines v3.0 and the 2025 design and setting data sheets as the required model-package framework. It emphasizes accurate representation of the physical arrangement and performance over expected conditions. A submitted droop number therefore needs supporting models and documentation; a fitted static curve alone does not demonstrate the required plant behavior.[8]

8 AEMO, Modelling requirements, Living page; lists guidelines dated 25 September 2025. Current model-package framework and Power System Model Guidelines v3.0.

8 Normal regulation fault response and recovery

Normal Volt/VAR droop specifies behavior around an operating voltage schedule. Fault response may use different voltage measurements, reactive-current laws, sequence controls, current priorities and controller states. Design and document the entry threshold, exit threshold, reset criteria and transitions. A high normal-operation gain does not by itself establish effective fault support.

During a deep dip, a constant-Q request can demand increasing current just as the converter reaches its limit. After clearing, residual integral states or delayed reactive commands can drive overvoltage. Test integrator tracking or freezing, current-limit release, active-power restoration, and the return to the normal curve together. Include asymmetric faults when negative-sequence behavior is relevant.

Separate command rate limits from physical response limits. A slow supervisory setpoint ramp may coexist with fast response to local voltage disturbance; applying the same ramp limiter indiscriminately can suppress needed support. Record which signal is limited and in which mode. Likewise, do not evaluate rise time against an unlimited command if the unit is physically saturated: identify the feasible final target and apply the governing test definition.

9 Grid requirements and their interpretation

9.1 Australia

The accessed NER S5.2.5.13 text allows voltage regulation at the connection point or another agreed location, with a setpoint adjusted for agreed droop or reactive-current compensation. Its automatic access provision includes regulation within 2% of that setpoint and configurable voltage reference capability. It also addresses coordination with other voltage controls and the nomination of highest and typical system impedances for tuning and compliance assessment. These provisions make measurement location, adjusted target and network envelope material parts of the design.[9]

The applicable registered performance standard and connection arrangements determine the project-specific obligations. The 2% regulation tolerance is not a recommendation for a 2% droop slope or deadband. Establish the applicable rules version and any transitional arrangements before a compliance assessment; a research comparison cannot determine the obligations of an unidentified existing plant.

9.2 European Union

Commission Regulation (EU) 2016/631, Article 21(3)(d), specifies a voltage-control slope capability range of at least 2–7% for the applicable power park modules, with increments no greater than 0.5%. The voltage-setpoint range is 0.95–1.05 pu in increments no greater than 0.01 pu. Article 2 defines slope using change from zero to maximum reactive power, making the normalization explicit. The prescribed ranges are capabilities, not an instruction that every plant operate at the same slope.[10]

The same article allows the relevant operator to select response times within specified ranges: 90% reactive response in 1–5 seconds and settling in 5–60 seconds, subject to its stated tolerance. National

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9 AEMC, National Electricity Rules S5.2.5.13 Voltage and reactive power control, Version-specific online text accessed 13 September 2026. Especially paragraphs (b)(2B), (i)–(j), and (m). Confirm project applicability and transitional rules.

10 European Commission, Commission Regulation EU 2016/631 establishing a network code on requirements for grid connection of generators, 14 April 2016; Official Journal 27 April 2016. Article 2(57) and Article 21(3)(d). National implementation and classification affect applicability.

implementation, module classification and connection-specific selections still matter. These times must not be transplanted to fault-current control or to a jurisdiction with a different test definition.

9.3 North America

NERC VAR-001-5 addresses maintaining voltage levels, reactive flows and resources within limits in real time. It establishes an operational context, not a universally optimal droop value. IEEE 2800-2022 supplies a transmission IBR performance framework, but its application and enforceability depend on adoption and the relevant arrangements. Distinguish a reliability standard, an IEEE technical standard and a NERC reliability guideline when describing a requirement.[11]

For procurement, request the applicable IEEE clauses and amendments in the compliance schedule rather than relying on a public summary. Exact IEEE numerical tuning and test requirements are not reproduced here because the accessed official catalogue provides scope rather than the complete normative clauses. Historical NERC and WECC documents remain useful technical sources but should not be treated as current software specifications or complete contemporary compliance checklists.

10 Worked curve design

Consider a plant at a 220 kV connection with Qbase = 100 MVAr, vref = 1.00, q0 = 0, dL = dH = 0.005, and KL = KH = 25. Let the assumed net limits be +60 MVAr injection and −40 MVAr absorption. These illustrative limits are treated as fixed for the calculation only; a real design would replace them with capability surfaces.

The injection limit is reached at v = 1.00 − 0.005 − 0.60/25 = 0.971 pu. The absorption limit is reached at v = 1.00 + 0.005 + 0.40/25 = 1.021 pu. The unequal saturation voltages result from asymmetric capability, despite equal slopes and deadbands.

Voltage pu	Voltage kV	Raw command MVAr	Limited command MVAr
0.965	212.3	+75.0	+60.0
0.975	214.5	+50.0	+50.0
0.990	217.8	+12.5	+12.5
1.000	220.0	0.0	0.0
1.010	222.2	−12.5	−12.5
1.025	225.5	−50.0	−40.0

This table evaluates the controller under imposed voltage values. It is not the closed-loop network trajectory. To calculate a connected equilibrium, couple the curve to the network and verify the resulting active segment. For example, take an uncontrolled network voltage vb = 0.98 pu and sensitivity H = 0.05 pu voltage per pu Q. In the injection segment q = 25(0.995 − v), and the network equation is v = 0.98 + 0.05q.

11 NERC, VAR-001-5 Voltage and Reactive Control, Effective 1 January 2019, per official status page. Official purpose and status. Operational framework rather than a tuning prescription.

Solving gives q = 0.1667 pu, or 16.67 MVAr, and v = 0.98833 pu. This voltage lies below 0.995 pu and the output is below 60 MVAr, so the assumed segment is consistent. Evaluating the curve at the uncontrolled 0.98 pu would command 37.5 MVAr, but that is not the equilibrium because the injection changes the voltage.

For a stronger disturbance with vb = 0.94 pu, the unsaturated solution is q = 0.6111 pu, exceeding the assumed 0.60 pu limit. With saturation, v = 0.94 + 0.05 × 0.60 = 0.97 pu. At this voltage the raw command is 0.625 pu, confirming that the saturated branch is consistent. This active-region check is essential in both hand calculations and optimization models.

11 A practical design and tuning procedure

11.1 Establish objectives and scenarios

First define the voltage-control location, allowed normal and post-contingency voltages, response criteria, reactive reserve targets, and intended participation with neighboring resources. Obtain the connection obligations and equipment capability maps. Build scenarios spanning high and low demand, renewable production, battery charging and discharging, seasonal topology, relevant outages and resource availability.

Record uncertainty explicitly: network impedance, load composition, measurement bias, controller delay, and unavailable equipment. Do not choose a single worst case solely by minimum short-circuit level. Different cases can determine voltage feasibility, dynamic damping, thermal capability and overvoltage recovery.

11.2 Select the initial curve

Choose the reactive normalization and physical gain first, then report percent droop. Locate the reference and bias so that expected operation leaves the required directional reserve. Choose a deadband that is compatible with measurement performance and support objectives. Use the saturation-voltage equations to check whether the resource reaches its needed capability before the relevant voltage boundary.

Coordinate common-bus participation through physical gains and availability. For distinct buses, use sensitivity analysis and AC studies to quantify both local benefit and remote effects. Check any compensation against transformer and network models. Reject candidates requiring infeasible internal voltages or unavailable reactive output.

11.3 Tune dynamics and verify robustness

With curve geometry fixed provisionally, tune filters, PI gains, unit allocation and communication treatment. Examine interacting modes across the scenario set, including each important limiter state. Select gains against explicit damping and response objectives; report the sensitivity of margins to model uncertainty. Revisit the static curve if acceptable dynamics cannot be achieved with the required gain.

Test large disturbances and recovery after the small-signal checks. Include meter failure and restoration, mode transfers, unit disconnection, delayed commands and stale measurements. Confirm that an unavailable or saturated resource does not leave an integrator accumulating an impossible request. Revalidate the final coupled design after any material change to either curve or dynamic settings.

11.4 Study acceptance matrix

Study	Principal result	Failure to detect
AC power flow and N minus 1	Voltages flows Q limits and reserve	Optimistic constant-voltage or unlimited-Q assumptions
Linearization and frequency response	Modes damping and control margins	Interaction hidden by single-plant tuning
Positive-sequence dynamics	Broad disturbance and dispatch coverage	Slow recovery or excessive overshoot
Detailed EMT	Fast interactions limits and unbalance	PLL current-limit or waveform instability
Controller integration tests	Real logic latency and mode transitions	Wrong sign stale data or integrator mismatch
Site commissioning	Measured curve and dynamic response	Model settings differ from installed controls

12 Droop in power flow and optimal power flow

12.1 Consistent equilibrium representation

An ideal PV bus imposes voltage until its reactive limit is reached. A droop-controlled bus instead imposes a relation between voltage and reactive output. In the active linear segment, include $v + r(q - q0) - vref = 0$ together with the network balance equations. Inside a fixed-output deadband use $q = q0$; at saturation use the binding reactive limit. Do not simultaneously impose an independent ideal voltage target at the same controller.

NERC's 2018 model-verification guide presents a plant with measured 4% droop for which a zero-droop power-flow representation overestimated reactive contribution. This is evidence of the consequence of a mismatched equilibrium model. Its discussion of software limitations is historical; present-day support must be checked for the particular tool and model.[12]

For a smooth nonlinear solver, a differentiable approximation can help convergence near the corners. Quantify its error around deadband edges and limits, and verify final candidates using the intended exact characteristic. Piecewise formulations, complementarity or mixed-integer approaches can represent active regions explicitly, but each introduces numerical or computational considerations. Maintain identical bases and measurement mappings between static and dynamic models.

12 NERC, Power Plant Model Verification for Inverter-Based Resources, September 2018. Chapter 1, printed page 3, Figure 1.2. Historical modelling example.

12.2 A proposed robust optimization structure

Let design variables include r, deadband widths, vref, q0 and optionally approved controller schedules. For each scenario s, enforce AC power balance, equipment limits, the droop relation and voltage constraints. A candidate objective can combine voltage deviation, losses, reserve shortfall and control effort:

$$minimize \ \Sigma s \ ws \ [\ av \ ||vs - vtarget,s||^2 + al \ Ploss,s + ar \ Rshort,s + aq \ ||qs - q0||^2 \ ]$$

Here ws are scenario weights and av, al, ar and aq are dimensionally consistent design weights. Use hard constraints for mandatory limits; the penalty form is for additional preferences and explicitly allowed shortfalls. Set critical contingency criteria independently of probability weights so that a low scenario weight cannot trade away required security.

This formulation is a proposed engineering structure, not a claim of convexity or global solvability. Optimizing r introduces products between settings and operating variables, while saturation introduces changing regions. Static constraints alone do not certify converter stability. A practical workflow screens or constrains candidates with validated dynamic surrogates, then subjects finalists to full dynamic and EMT verification.

12.3 Transfer from published optimization research

Murzakhanov and coauthors optimize deadband-and-saturation Volt/VAR rules for a single-phase distribution feeder using anticipated scenarios and a stability-constrained parameterization. Their 141-bus feeder study demonstrates a useful approach to tuning complete curves rather than slopes alone. The reported guarantees depend on their feeder and controller model; transferring them to meshed transmission with PLLs, excitation dynamics and current limits requires new modelling and validation.[13]

For transmission research, useful comparisons include fixed uniform droop, location-dependent fixed droop, scheduled intercepts, jointly optimized curves, and coordinated secondary control. Report voltage security, directional reserve, damping, losses, control movement and computational burden. Use unseen contingencies and parameter perturbations for evaluation, rather than evaluating only on the scenarios used to choose settings.

13 Commissioning operation and observed failure

13.1 Commissioning measurements

Record synchronized voltage, active and reactive power at the agreed point; relevant unit measurements; reference commands; active control mode; capability limits; integrator or limiter status; tap and shunt states; and communication health. Sample fast enough for the dynamics under assessment. Establish instrument scaling, sign and time alignment before fitting a curve.

Test the deadband edges, both slope directions, relevant references and capability transitions using approved perturbations. Estimate the characteristic from settled operating points within a consistent mode. A regression through arbitrary operational data can confuse changing schedules, network feedback and saturation with the implemented droop. For dynamic verification, compare measured and simulated responses under matched initial conditions.

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13 Ilgiz Murzakhanov, Sarthak Gupta, Spyros Chatzivasileiadis and Vassilis Kekatos, Optimal Design of Volt/VAR Control Rules for Inverter-Interfaced Distributed Energy Resources, 23 May 2023, arXiv v2. Sections II–V. Distribution-feeder assumptions and 141-bus evaluation.

Specify acceptance in measurable terms: curve slope and intercept tolerance, onset of response, rise and settling definitions, permitted overshoot, damping, and absence of sustained oscillation or limit cycling. The values should come from the governing project criteria. Keep an approved settings register that maps each physical parameter to the controller and model parameter names, firmware versions and evidence files.

13.2 South Australian incident

AEMO's February 2023 report on the 23 June 2022 South Australian incident attributes oscillations originating at Port Augusta Renewable Energy Park to incorrect control code introduced during a communication fail-safe test. Reactive oscillations reached 247 MVAr peak-to-peak and persisted for almost five hours; AEMO reported that transmission voltages remained within relevant limits. The report's fail-safe discussion describes a failure to return correctly to the voltage droop loop.[14]

The lesson is that a valid curve specification is insufficient if switching logic does not restore it. A suitable commissioning plan should include verification of final code and settings after each test, confirmation of the active control path after communication recovery, and continued monitoring after the test is declared complete. This event does not establish that a particular numerical slope was the root cause.

13.3 Operating triggers for reassessment

Reassess the design after material changes to network topology, nearby resource controls, plant capacity, compensation settings or controller firmware. Monitor sustained reactive circulation, operation close to directional limits, repeated deadband crossings, tap hunting and growing oscillatory components. Persistent behavior should trigger model-based diagnosis, not repeated manual slope changes without a record of the underlying cause.

14 Research priorities and design conclusions

The strongest research opportunity is a transmission-specific co-design of droop geometry, capability and dynamic margins under uncertainty. A useful formulation would preserve permanent droop in normal operation, explicitly manage current-limited recovery, and account for both local and remote voltage constraints. Its value should be measured against robust fixed settings, not only against an untuned baseline.

Several questions remain practically important. How much gain adaptation is justified by changing impedance when estimation error is included? Can reactive reserve be allocated economically without concentrating dynamic duty on the electrically nearest resource? What stability certificate survives limit activation and transfers between GFL, GFM and synchronous control behavior? How can aggregate models retain the internal constraint that determines the actual connection-point capability?

Wang and coauthors investigate dispatching GFM sources by changing droop intercepts, with demonstrations in a mixed-source microgrid. That work supports treating dispatch and primary droop as related design variables. It does not establish the suitability of unrestricted intercept changes for a large transmission plant; the research extension should assess ramping, coordination and dynamic limits at the transmission scale.[15]

For an implementable design, deliver the full curve, its bases and sign conventions, the capability surfaces, control diagrams, approved modes, tested network envelope, model mappings and commissioning evidence.

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14 AEMO, Power System Oscillations in South Australia on 23 June 2022, February 2023. Overview, printed page 6, and Appendix A2, printed page 19.

15 Jing Wang, Subhankar Ganguly, Soham Chakraborty and Benjamin Kroposki, Dispatching Grid-Forming Inverters in Grid-Connected and Islanded Mode Preprint, 2024. 2024 IEEE PES General Meeting. Droop-intercept dispatch concept and microgrid demonstration.

Select the numerical slope only after these elements agree. The intended result is a predictable allocation of reactive support with acceptable voltage performance and demonstrated dynamic behavior throughout the approved operating envelope.

Appendix A Parameter specification checklist

Parameter group	Minimum information to record
Measurement	Bus and voltage level sequence definition polarity filtering and latency
Normalization	Vbase Qbase Sbase current base and percent-droop convention
Curve	Reference bias both slopes both deadbands and saturation treatment
Capability	Q limits versus P V availability and thermal or DC conditions
Control	PI parameters filters sample times anti-windup and allocation logic
Coordination	Remote compensation taps shunts neighboring controls and secondary regulation
Modes	Normal fault current-limited fallback and recovery states
Validation	Scenarios models firmware versions acceptance criteria and measured evidence

Appendix B Common mistakes and corrections

Mistake	Correction
Comparing 4% and 5% without bases	Convert both to MVAr per kV at the same location
Calling lower droop a lower gain	Use K = 1/r and label axes explicitly
Applying the curve to unchanged network voltage	Solve the curve and AC network simultaneously
Assuming nameplate Q remains available during a dip	Use voltage-dependent current and internal limits
Using ideal PV control in a droop study	Enforce the droop equation and active limits
Treating line compensation as ordinary positive droop	Verify effective polarity with a perturbation

Mistake	Correction
Using a feeder convergence bound as an EMT guarantee	Match the stability claim to the actual model
Validating only normal mode	Test fault limits communication failure and restoration

Sources

The sources below were accessed on 13 September 2026. Publication dates are those stated by the documents or their publishers. Numbered footnotes identify the sources supporting the associated passages. Calculations and proposed workflows are engineering analysis under the assumptions stated in the text. The regulatory comparison identifies source provisions and does not determine connection-specific applicability.

1. NERC. Inverter-Based Resource Performance Guideline. September 2018. Chapter 3, especially printed page 28. Historical technical guideline.

2. IEEE Standards Association. IEEE 2800-2022 Standard for Interconnection and Interoperability of Inverter-Based Resources Interconnecting with Associated Transmission Electric Power Systems. 22 April 2022. Official catalogue scope and status; full normative text is not reproduced.

3. Hitachi Energy. STATCOM. Undated. Manufacturer description of VSC reactive compensation and shunt combinations.

4. Babak Abdolmaleki, John W. Simpson-Porco and Gilbert Bergna-Diaz. Distributed Optimization for Reactive Power Sharing and Stability of Inverter-Based Resources Under Voltage Limits. 28 February 2023, arXiv v2. Sections II–V. Microgrid demonstrations; transmission transfer requires validation.

5. WECC. Central Station Photovoltaic Power Plant Model Validation Guideline. 17 June 2015. Section 3.2.1. Historical REPC_A mode and compensation guidance.

6. AEMC. Generator Technical Performance Standards Final Determination. 27 September 2018. Printed page 131, droop and reactive current compensation.

7. Yashen Lin and coauthors, NREL. Research Roadmap on Grid-Forming Inverters. November 2020. NREL/TP-5D00-73476; voltage-control discussion, printed page 19.

8. AEMO. Modelling requirements. Living page; lists guidelines dated 25 September 2025. Current model-package framework and Power System Model Guidelines v3.0.

9. AEMC. National Electricity Rules S5.2.5.13 Voltage and reactive power control. Version-specific online text accessed 13 September 2026. Especially paragraphs (b)(2B), (i)–(j), and (m). Confirm project applicability and transitional rules.

10. European Commission. Commission Regulation EU 2016/631 establishing a network code on requirements for grid connection of generators. 14 April 2016; Official Journal 27 April 2016. Article 2(57) and Article 21(3)(d). National implementation and classification affect applicability.

11. NERC. VAR-001-5 Voltage and Reactive Control. Effective 1 January 2019, per official status page. Official purpose and status. Operational framework rather than a tuning prescription.

12. NERC. Power Plant Model Verification for Inverter-Based Resources. September 2018. Chapter 1, printed page 3, Figure 1.2. Historical modelling example.

13. Ilgiz Murzakhanov, Sarthak Gupta, Spyros Chatzivasileiadis and Vassilis Kekatos. Optimal Design of Volt/VAR Control Rules for Inverter-Interfaced Distributed Energy Resources. 23 May 2023, arXiv v2. Sections II–V. Distribution-feeder assumptions and 141-bus evaluation.

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15. Jing Wang, Subhankar Ganguly, Soham Chakraborty and Benjamin Kroposki. Dispatching Grid-Forming Inverters in Grid-Connected and Islanded Mode Preprint. 2024. 2024 IEEE PES General Meeting. Droop-intercept dispatch concept and microgrid demonstration.